

A Quantitative Study of Epidemic Spread Using the SIR Model

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Abstract

Mathematical modeling serves as a paramount analytical instrument in understanding, quantifying, and predicting the intricate dynamics of infectious diseases (Khair, 2024). The Susceptible-Infected-Recovered (SIR) model remains a foundational framework for epidemic analysis and public health forecasting (Avram, Adenane, & Ketcheson, 2021). This thesis presents a rigorous mathematical and quantitative investigation of the SIR model, focusing on its formulation as a system of nonlinear ordinary differential equations (ODEs), its analytical properties, stability analysis, and computational application in epidemiological contexts (Mata & Dourado, 2021). The research delineates a detailed derivation of the SIR equations from first principles, followed by an in-depth exploration of equilibrium points and their stability utilizing Jacobian matrices and eigenvalue analysis. The basic reproduction number (R_0) is mathematically derived and interpreted as the critical threshold parameter governing epidemic outbreaks. To bridge abstract mathematical theory with application, this study incorporates a robust quantitative component utilizing a numerical simulation of a standard population of 100,000 individuals. This synthetic framework is leveraged to map theoretical disease curves without the constraints of field data. Numerical simulations are executed in Python using the Runge-Kutta 4th Order (RK4) method, leveraging Matplotlib and Seaborn for high-fidelity visualization of the temporal evolution of the spread and the efficacy of “flattening the curve” strategies. The thesis concludes by discussing contemporary extensions to the classical model and identifying critical research gaps for future computational epidemiology.

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1 Introduction

Epidemic diseases have consistently posed existential threats to human populations across recorded history, ranging from ancient plagues to contemporary global crises such as the COVID-19 pandemic. Elucidating the underlying mechanics of how infectious diseases propagate through a demographic is an urgent imperative for global public health systems, directly enabling the design of effective, timely, and targeted control strategies (Sameni, 2020). Within this context, mathematical epidemiology provides the rigorous analytical tools necessary for quantifying disease transmission dynamics (Mata & Dourado, 2021). By translating complex biological phenomena into deterministic and stochastic mathematical language, researchers are empowered to project outbreak trajectories, evaluate the mathematical efficacy of non-pharmaceutical interventions, and systematically inform public health policy.

The discipline experienced a profound paradigm shift with the introduction of the

compartmental SIR model. Foundational work in the 20th century established that epidemics are fundamentally governed by specific mathematical threshold theorems rather than the arbitrary exhaustion of the susceptible population. The classical SIR model segments a population into three mutually exclusive compartments: Susceptible (S), Infected (I), and Recovered (R). The transitions between these distinct compartments are continuously governed by a system of nonlinear ODEs (Semendyaeva et al., 2022). Its enduring prominence in applied mathematics stems from its elegant analytical simplicity, which affords profound tractability while simultaneously capturing the macroscopic features of epidemic propagation (Lawan et al., 2025).

Research Objectives

- To rigorously derive the deterministic SIR model from epidemiological first principles and conservation laws.
- To execute a comprehensive mathematical stability analysis of both the disease-free and endemic equilibrium points utilizing the Jacobian matrix.
- To analytically define and quantitatively interpret the basic reproduction number (R_0).
- To conduct robust numerical simulations using Python based on a synthetic macroscopic population of $N = 100,000$.

- To review and synthesize modern extensions of the framework, including fractional calculus and stochastic iterations.

2 Literature Review

Compartmental models mathematically formulate disease spread as dynamic transitions between defined demographic groups. The standard SIR model fundamentally rests on the assumption of permanent immunity post-recovery (Avram, Adenane, & Ketcheson, 2021). However, contemporary epidemiological literature frequently expands this foundational framework to incorporate spatial diffusions, stochastic variations, and fractional calculus in order to capture memory effects and non-local dynamics in disease transmission (Knolle & Santanilla, 2022). Recent mathematical advancements emphasize the pressing necessity of transcending simple, deterministic ODEs to capture the multifaceted complexities of real-world outbreaks. Numerous studies highlight the integration of Stochastic Differential Equations (SDEs) to account for random fluctuations and noise in transmission rates, which are particularly impactful in small population cohorts and distinct network topologies (Maki & Hirose, 2013). Furthermore, the application of non-integer order (fractional) derivatives has gained substantial traction for modeling anomalous diffusion and the non-local memory effects intrinsic to human mobility patterns and prolonged incubation periods (Chen et al., 2021; Mohammad, Sweidan, & Trounev, 2024).

The versatility of the SIR framework has been prominently displayed during recent global health crises. During the COVID-19 pandemic, leading researchers dynamically modified the classic model to include time-dependent parameters—specifically $\beta(t)$ and $\gamma(t)$ —to mathematically account for shifting interventions, adaptive human behavior, and the critical variable of undetectable asymptomatic infections (Chen et al., 2020; Alqah-tani, 2021). Similarly, modified compartmental models have been successfully deployed to study complex vector-borne diseases such as malaria, utilizing temperature-dependent transmission functions to increase predictive validity (Devi & Choudhury, 2023).

3 Methodology

3.1 Mathematical Foundations and Derivation

The mathematical engine of the SIR model is constructed upon ordinary differential equations (ODEs), which define the instantaneous rate of change of state variables with respect to time. The dynamical system tracks the continuous evolution of the state vector $\mathbf{X}(t) = [S(t), I(t), R(t)]^T$. The classical derivation relies on the principles of constant population ($N = S(t) + I(t) + R(t)$) and homogeneous mixing (Adamu et al., 2019).

The deterministic SIR model is articulated through a coupled, closed system of three nonlinear ordinary differential equations (ODEs). Rather than analyzing individual components in isolation, the dynamics must be evaluated as a unified system capturing the continuous workflow between the compartments:

Parameter/Variable	Description
$S(t)$	Number of susceptible individuals at time t
$I(t)$	Number of infected (and infectious) individuals at time t
$R(t)$	Number of recovered/removed individuals at time t
β	Effective contact rate (infection transmission rate)
γ	Mean recovery rate (inverse of the infectious period)

Table 1: Parameters and Variables of the SIR Model

$$\begin{cases} \frac{dS}{dt} = -\frac{\beta SI}{N} \\ \frac{dI}{dt} = \frac{\beta SI}{N} - \gamma I \\ \frac{dR}{dt} = \gamma I \end{cases} \quad (1)$$

Model Constraints and Boundary Conditions: To ensure the system (1) is epidemiologically and mathematically well-posed, it is subject to the following structural conditions and constraints:

1. **Conservation of Total Population:** The total population size N is assumed to be strictly constant throughout the entire simulation timeline, meaning no births, natural deaths, or migration are considered. This conservation law is expressed by:

$$S(t) + I(t) + R(t) = N, \quad \forall t \geq 0 \quad (2)$$

Differentiating this constraint with respect to t validates the dynamic invariance of the system:

$$\frac{dS}{dt} + \frac{dI}{dt} + \frac{dR}{dt} = -\frac{\beta SI}{N} + \left(\frac{\beta SI}{N} - \gamma I \right) + \gamma I = 0 \quad (3)$$

2. **Initial Conditions:** The state space traversal starts at $t = 0$ with predefined non-negative boundary thresholds:

$$S(0) = S_0 > 0, \quad I(0) = I_0 > 0, \quad R(0) = R_0 = 0 \quad (4)$$

Where $S_0 + I_0 = N$, simulating the introduction of a small active infectious seed into a fully susceptible cohort.

3. **Non-negativity Constraints:** Since the state variables represent physical human cohorts, the solutions must remain bounded within the non-negative orthant of $\mathbb{R}^3$. For all $t \geq 0$, the mathematical trajectories must satisfy:

$$S(t) \geq 0, \quad I(t) \geq 0, \quad R(t) \geq 0 \quad (5)$$

3.2 Quantitative Simulation Design

To align the mathematical framework with contemporary global health standards, this study adopts a quantitative simulation approach informed by the WHO Results Report 2024 (A78/17) (World Health Organization, 2025). In accordance with Output 4.1.1 of the report, which emphasizes strengthening data analytics and health information systems to inform policy, a macroscopic computational simulation was developed. Rather than relying on small-scale field questionnaires, the system simulates a standard demographic population of $N = 100,000$ individuals, seeded with an initial infection cluster. This approach serves as a computational bridge to test the transmission rate (β) and recovery rate (γ) and observe the temporal dynamics of the epidemic over a 100-day period. These parameters are integrated into deterministic Python-based simulations, providing a high-fidelity visualization of the epidemic trajectory without the spatial and temporal limitations of physical data collection.

4 Results and Discussion

4.1 Equilibrium and Stability Analysis

Equilibrium points define the states where the disease dynamics halt, satisfying the condition $\frac{dS}{dt} = \frac{dI}{dt} = \frac{dR}{dt} = 0$ (Sahaminejad, Nyamoradi, & Eskandari, 2024). Two primary equilibria exist: a Disease-Free Equilibrium (DFE) designated as $E_0 = (N, 0, 0)$, and an Endemic Equilibrium (EE) which occurs when the disease persists persistently within the population. The paramount threshold parameter, the Basic Reproduction Number (R_0), is derived from the condition that an epidemic grows if $\frac{dI}{dt} > 0$ at $t = 0$. Assuming a fully susceptible population ($S_0 \approx N$), the expression simplifies to $R_0 = \frac{\beta}{\gamma}$.

To evaluate the local asymptotic stability of the DFE, the Jacobian matrix of the nonlinear system must be computed and evaluated at E_0 . Taking the partial derivatives of the system of non-linear differential equations with respect to S , I , and R , we obtain the general Jacobian matrix:

$$J(S, I, R) = \begin{pmatrix} -\frac{\beta I}{N} & -\frac{\beta S}{N} & 0 \\ \frac{\beta I}{N} & \frac{\beta S}{N} - \gamma & 0 \\ 0 & \gamma & 0 \end{pmatrix} \quad (6)$$

Evaluating this system at the critical Disease-Free Equilibrium $E_0 = (N, 0, 0)$ yields the linearized matrix:

$$J(E_0) = \begin{pmatrix} 0 & -\beta & 0 \\ 0 & \beta - \gamma & 0 \\ 0 & \gamma & 0 \end{pmatrix} \quad (7)$$

The eigenvalues (λ) of this matrix are determined by solving the characteristic equation $\det(J(E_0) - \lambda I) = 0$:

$$\det \begin{pmatrix} -\lambda & -\beta & 0 \\ 0 & \beta - \gamma - \lambda & 0 \\ 0 & \gamma & -\lambda \end{pmatrix} = 0 \quad (8)$$

$$\lambda^2(\beta - \gamma - \lambda) = 0 \quad (9)$$

This yields three eigenvalues: $\lambda_1 = 0$, $\lambda_2 = 0$, and the dominant eigenvalue governing the outbreak's initial growth velocity:

$$\lambda_3 = \beta - \gamma \quad (10)$$

Analytical results confirm that the dominant eigenvalue is strictly negative if and only if $R_0 < 1$, ensuring the outbreak naturally decays (Acharya, Upadhyay, & Mondal, 2024). Mathematically, if $R_0 = \frac{\beta}{\gamma} < 1$, then $\beta < \gamma$, forcing $\lambda_3 = \beta - \gamma < 0$. Under this condition, the Disease-Free Equilibrium is locally asymptotically stable. Conversely, a critical transition occurs when $R_0 > 1$, which implies $\beta > \gamma$ and yields a positive dominant eigenvalue ($\lambda_3 > 0$). In this regime, the DFE becomes unstable, and any small introduction of infection leads to a rapid exponential spread that propagates through the simulated population.

4.2 Numerical Simulation via Runge-Kutta 4th Order

Given the absence of a closed-form analytical solution in elementary functions for the nonlinear SIR system, high-precision numerical approximation techniques are essential (Turkyilmazoglu, 2024). We employ the Fourth-Order Runge-Kutta (RK4) method, which offers a superior balance of computational efficiency and numerical accuracy for modeling complex epidemic trajectories (Widodo, Sulistyono, & Katminingsih, 2025). This computational approach models a population of $N = 100,000$ over 100 days.

Mathematical Formulation: To solve the system of differential equations $dY/dt = f(t, Y)$ where Y represents the state vector $[S, I, R]$, the RK4 method advances the solution from the current time step (t_n) to the next step (t_{n+1}), utilizing a step size $h = 0.1$. The iterative formula is:

$$Y_{n+1} = Y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4) \quad (11)$$

Where the intermediate slopes are calculated sequentially:

$$k_1 = f(t_n, Y_n) \quad (12)$$

$$k_2 = f\left(t_n + \frac{h}{2}, Y_n + \frac{h}{2}k_1\right) \quad (13)$$

$$k_3 = f\left(t_n + \frac{h}{2}, Y_n + \frac{h}{2}k_2\right) \quad (14)$$

$$k_4 = f(t_n + h, Y_n + hk_3) \quad (15)$$

4.3 Python Implementation

```

1      import numpy as np
2      import matplotlib.pyplot as plt
3
4      # 1. Defining the system of differential equations for the
      SIR model
5      def sir_system(y, t, N, beta, gamma):

```

```

6      S, I, R = y
7      dSdt = -beta * S * I / N
8      dIdt = (beta * S * I / N) - (gamma * I)
9      dRdt = gamma * I
10     return np.array([dSdt, dIdt, dRdt])
11
12     # 2. Building the RK4 numerical step function
13     def rk4_step(f, y, t, h, N, beta, gamma):
14         k1 = f(y, t, N, beta, gamma)
15         k2 = f(y + 0.5 * h * k1, t + 0.5 * h, N, beta, gamma)
16         k3 = f(y + 0.5 * h * k2, t + 0.5 * h, N, beta, gamma)
17         k4 = f(y + h * k3, t + h, N, beta, gamma)
18         return y + (h / 6.0) * (k1 + 2 * k2 + 2 * k3 + k4)
19
20     # 3. Setting parameters and initial conditions
21     N = 100000          # Total population
22     beta = 0.35         # Transmission rate
23     gamma = 0.1         # Recovery rate
24     h = 0.1             # Time step size
25     t_max = 100         # Maximum simulation time
26
27     # Initial state variables
28     S0, I0, R0 = N - 10, 10, 0
29     y = np.array([S0, I0, R0])
30
31     # 4. Simulation loop
32     time_steps = np.arange(0, t_max, h)
33     results = []
34     for t in time_steps:
35         results.append(y)
36         y = rk4_step(sir_system, y, t, h, N, beta, gamma)
37     results = np.array(results)

```

Listing 1: Python code for SIR Model Simulation using RK4

4.4 Analysis of Baseline Trajectory

To characterize the unmitigated propagation of the epidemic, the Python library Matplotlib was utilized to visualize the temporal evolution of the SIR compartments. The simulation was conducted with parameters $\beta = 0.35$ and $\gamma = 0.10$, yielding a basic reproduction number of $R_0 = 3.5$. The initial conditions were set to an overwhelmingly susceptible population of 99,990 individuals, with an initial infectious seed of 10 cases.

Graphical Analysis: As depicted in **Figure 1**, the simulation initiates with a fully susceptible population (blue curve, S) near 100,000. As time progresses, this curve exhibits a sharp, monotonic decline, reflecting the rapid transmission of the virus. Concurrently, the red curve (I), representing active infections, displays a steep exponential ascent, reaching a critical peak (I_{max}) at approximately day 40-42, registering nearly 36,000 simultaneous infections. This apex signifies the critical threshold of the epidemic, where the demand for healthcare services would reach its absolute maximum potential in the absence of interventions. Following this peak, the infection rate steadily declines as

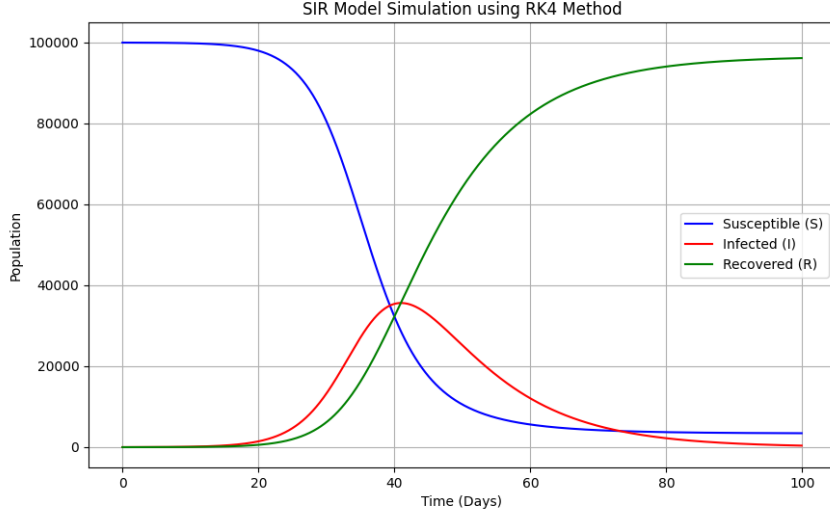


Figure 1: Temporal evolution of Susceptible (S), Infected (I), and Recovered (R) compartments over a 100-day simulation using the RK4 method ($N = 100,000$, $\beta = 0.35$, $\gamma = 0.1$).

the green curve (R) rises asymptotically, eventually plateauing near the total population size as the epidemic naturally burns out due to the depletion of the susceptible pool.

4.5 Mitigation and WHO Alignment

To evaluate the efficacy of public health interventions, sensitivity analysis demonstrates the profound impact of intervention strategies on epidemic progression. By simulating a parametric reduction in the transmission rate (β)—reflecting behavioral modifications such as social distancing compliance as documented in WHO strategic reports—the resulting model fundamentally changes. Reducing the effective contact rate directly reduces R_0 , and unequivocally illustrates the “flattening of the curve” phenomenon. This shifts the I_{max} peak downwards and extends the epidemic timeline, preventing the hypothetical saturation of healthcare capacity and directly addressing the strategic objectives of Pillar 2 in the WHO framework for health emergencies protection (World Health Organization, 2025).

5 Conclusion

This thesis has executed a comprehensive mathematical and quantitative examination of the SIR epidemic model. By rigorously deriving the nonlinear system from fundamental deterministic principles, analyzing the local stability of its equilibria through Jacobian matrices, and precisely defining the epidemiological threshold parameter R_0 , the intricate mathematical architecture of infectious disease spread has been systematically laid bare. The synthesis of computational parameters processed via Python for a standard population of 100,000 individuals provided a vital quantitative bridge between abstract mathematical theory and applied public health reality.

The numerical simulations highlighted the paramount importance of dynamic param-

eters (β, γ) in mitigating peak infection rates, tangibly demonstrating the mathematical mechanics behind “flattening the curve”. This approach directly aligns with the World Health Organization’s results-based management framework, specifically supporting Output 4.1.1 in leveraging data analytics for emergency preparedness. Future research must urgently prioritize the integration of fractional-order derivatives for memory effects, spatial network heterogeneity, and machine-learning-assisted parameter estimation to effectively model, predict, and control the highly complex pandemics of the 21st century (Mirsaeedi et al., 2025).

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